

$$\begin{aligned} \left(\frac{1}{5}A^{-1}\right)(5A) &= \frac{1}{5} \cdot (A^{-1} \cdot 5A) \text{ (Theorem 2(d))} \\ &= \frac{1}{5} \cdot 5(A^{-1}A) \text{ (Theorem 2(d))} \\ &= I. \end{aligned}$$

(Existence part proved)

$$\begin{aligned} \text{Let, there exist a matrix } X \text{ s.t. } (5A)X &= I \Rightarrow A^{-1} \cdot (5A \cdot X) = A^{-1}I \\ \Rightarrow (A^{-1} \cdot 5A)X &= A^{-1}I \text{ (Theorem 2(a))} \Rightarrow (5(A^{-1}A))X = A^{-1}I \text{ (Theorem 2(d))} \\ \Rightarrow (5I)X &= A^{-1}I \text{ (Given)} \Rightarrow 5(IX) = A^{-1}I \text{ (Theorem 2(d))} \\ \Rightarrow 5X &= A^{-1} \text{ (Theorem 2(e))} \Rightarrow \frac{1}{5} \cdot (5X) = \frac{1}{5} \cdot A^{-1} \Rightarrow \left(\frac{1}{5} \cdot 5\right)X = \frac{1}{5}A^{-1} \\ &\Rightarrow X = \frac{1}{5} \cdot A^{-1} \text{ (Theorem 1(f))} \end{aligned}$$

$$\begin{aligned} \text{Let, there exist a matrix } Y \text{ s.t. } Y(5A) &= I \Rightarrow (Y(5A)) \cdot A^{-1} = I \cdot A^{-1} \\ \Rightarrow Y \cdot (5A \cdot A^{-1}) &= I \cdot A^{-1} \text{ (Theorem 2(a))} \Rightarrow Y \cdot (5(AA^{-1})) = IA^{-1} \text{ (Theorem 2(d))} \\ \Rightarrow Y \cdot (5I) &= IA^{-1} \text{ (Given)} \Rightarrow 5(YI) = IA^{-1} \text{ (Theorem 2(d))} \\ \Rightarrow 5Y &= A^{-1} \text{ (Theorem 2(e))} \Rightarrow \frac{1}{5} \cdot (5Y) = \frac{1}{5} \cdot A^{-1} \Rightarrow \left(\frac{1}{5} \cdot 5\right)Y = \frac{1}{5} \cdot A^{-1} \\ &\Rightarrow Y = \frac{1}{5} \cdot A^{-1} \text{ (Theorem 1(f))} \end{aligned}$$

$\Rightarrow Y = \frac{1}{5} \cdot A^{-1}$ (Uniqueness part proved)

$\therefore$ There exists a unique matrix $\frac{1}{5}A^{-1}$ s.t. $(5A)$ on both left and right multiplication give I . By defn, $(5A)$ is invertible. $\square$